

Causal Modeling in R: Whole Game

Malcolm Barrett
Stanford University

- 1 Specify causal question (e.g. target trial)
- 2 Draw assumptions (causal diagram)
- 3 Model assumptions (e.g., propensity)
- 4 Diagnose model (e.g., balance)
- 5 Estimate causal effects (e.g., IPW)
- 6 Sensitivity analysis

We'll focus on the broader ideas behind each step and what they look like all together; we don't expect you to fully digest each idea. We'll spend the rest of the workshop taking up each step in detail

Does mosquito bed net use reduce malaria risk?

```
1 library(causalworkshop)
2 net_data
```

```
# A tibble: 1,752 × 10
```

	id	net	net_num	malaria_risk	income	health	household
	<int>	<lgl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	FALSE	0	38	779	35	1
2	2	FALSE	0	48	700	35	3
3	3	FALSE	0	32	1083	58	3
4	4	FALSE	0	55	753	68	3
5	5	FALSE	0	36	919	46	5
6	6	FALSE	0	30	969	37	3
7	7	FALSE	0	29	1012	58	1
8	8	FALSE	0	45	708	30	2
9	9	FALSE	0	51	733	18	3
10	10	FALSE	0	42	862	64	3

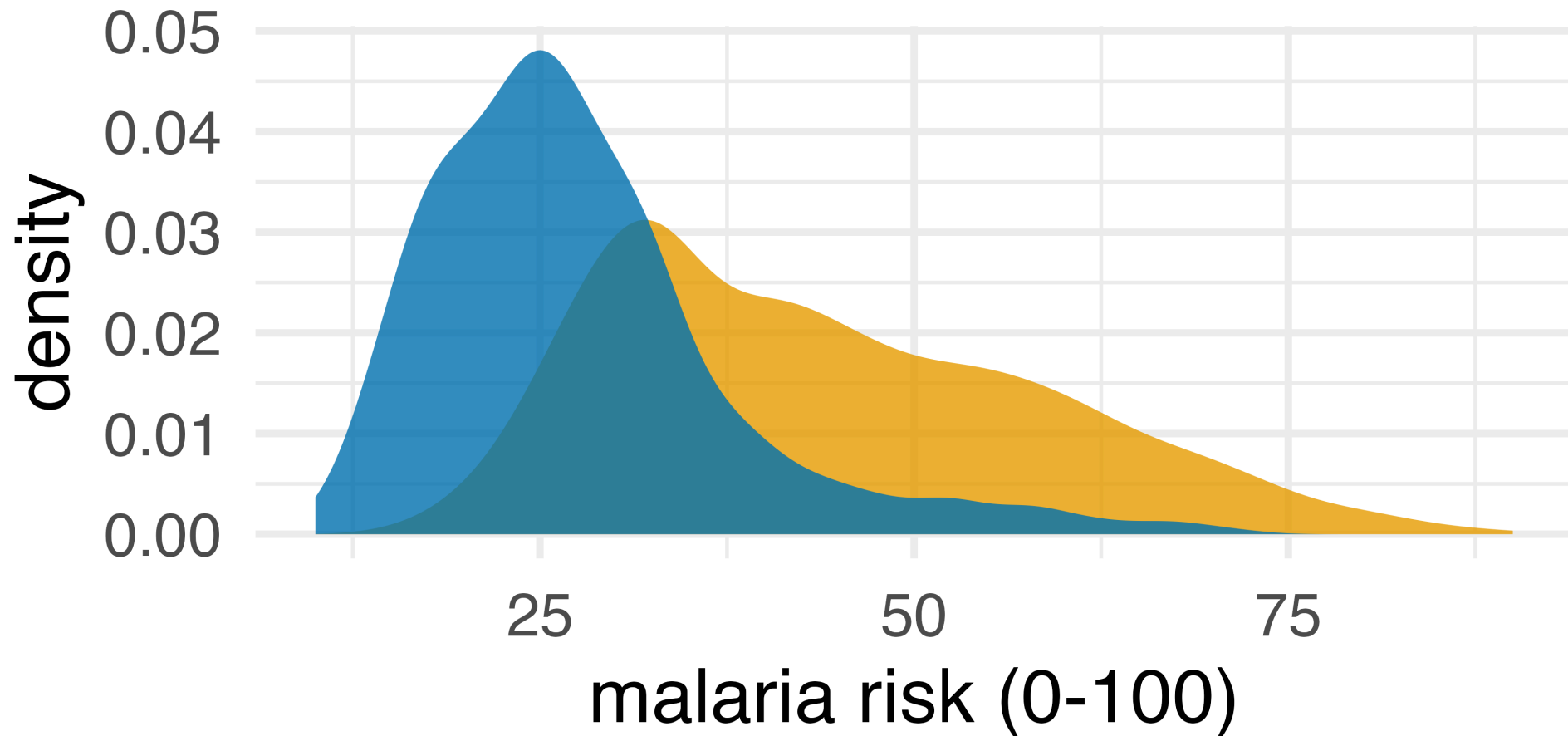
```
# i 1,742 more rows
```

```
# i 3 more variables: eligible <lgl>, temperature <dbl>,
```

```
# insecticide_resistance <dbl>
```

Thanks to Andrew Heiss for the data!

Do net users have lower malaria risk?



used a net FALSE TRUE

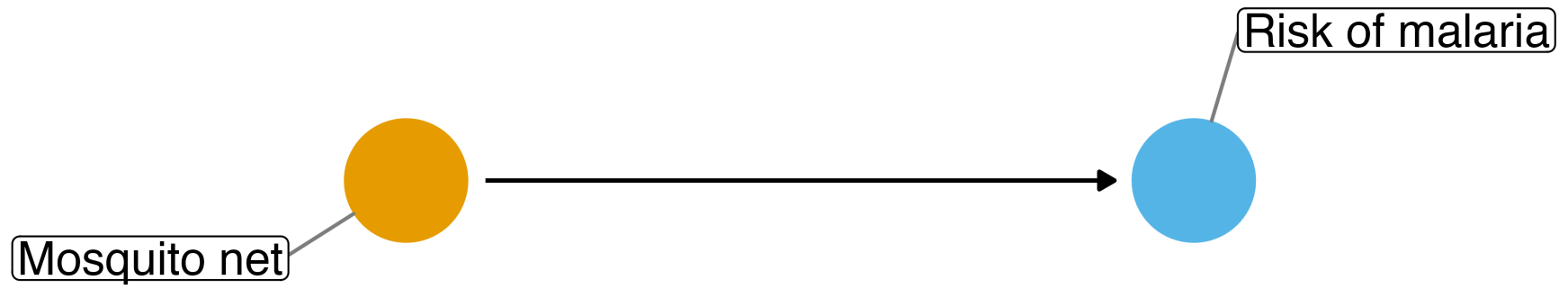
Do net users have lower malaria risk?

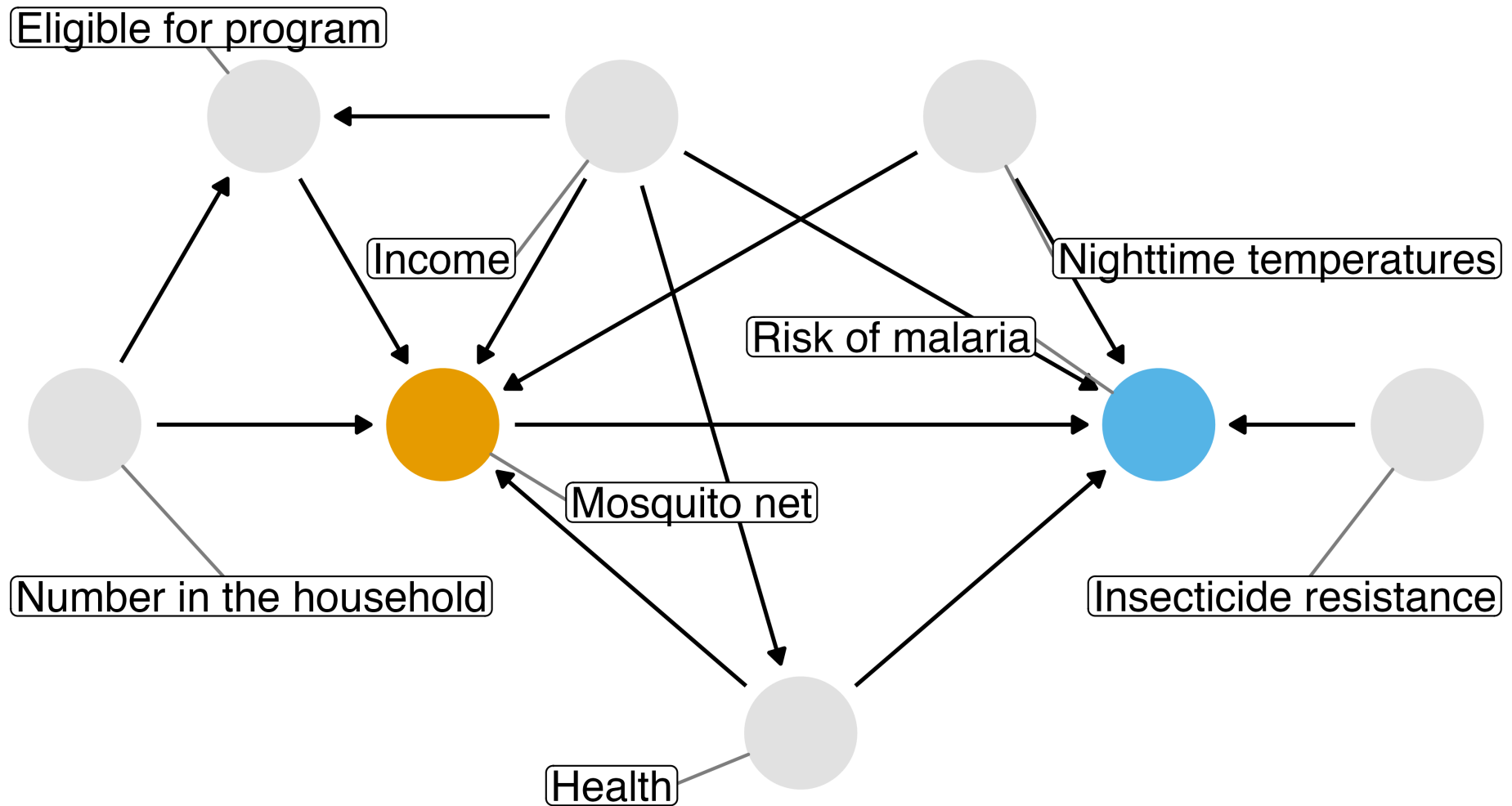
```
1 # ~16 points lower risk for net users vs. non-users
2 net_data |>
3   group_by(net) |>
4   summarize(
5     mean_malaria_risk = mean(malaria_risk),
6     sd = sd(malaria_risk),
7     .groups = "drop"
8   )
```

A tibble: 2 × 3

	net	mean_malaria_risk	sd
	<lgl>	<dbl>	<dbl>
1	FALSE	43.9	14.6
2	TRUE	27.5	10.3

draw your assumptions

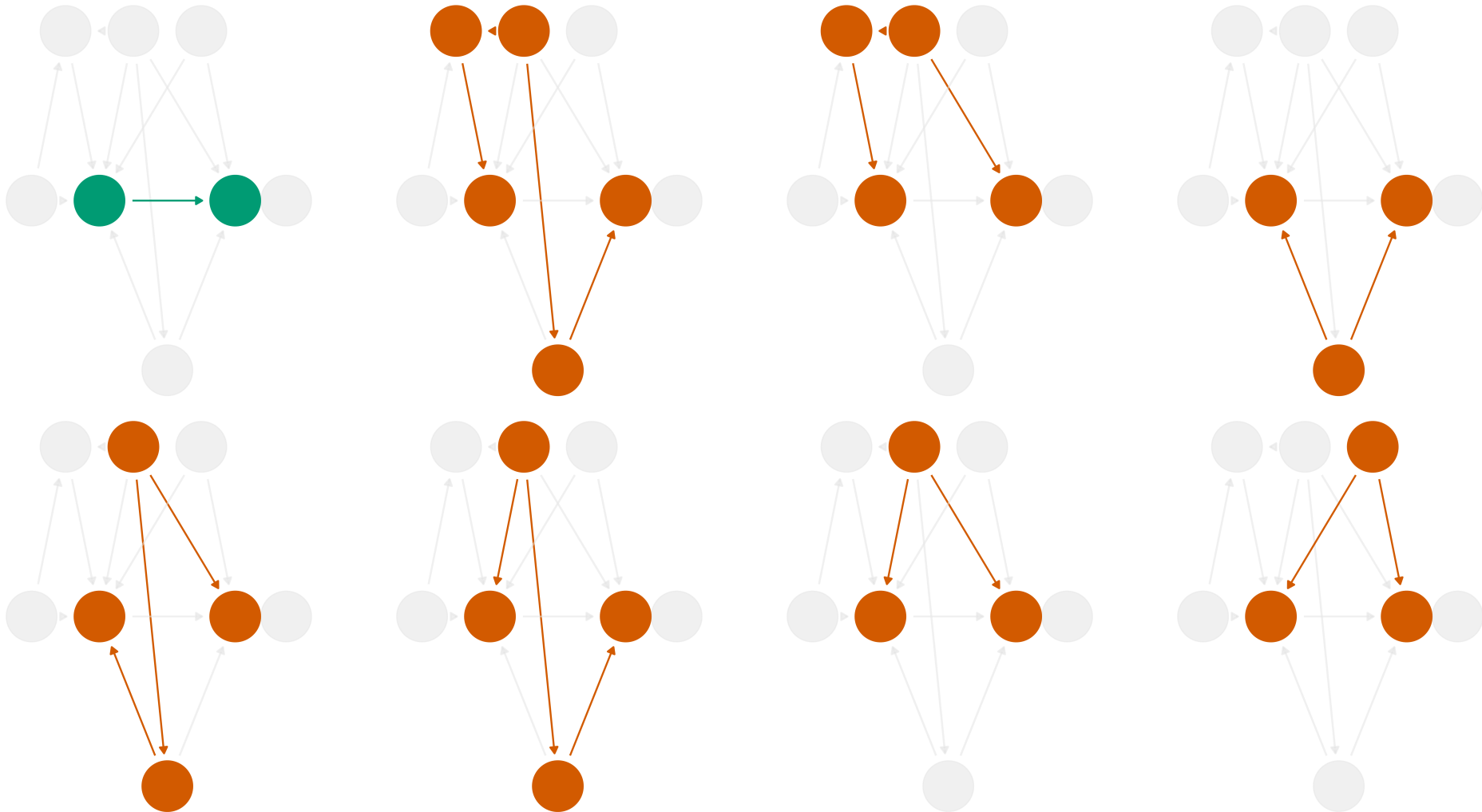




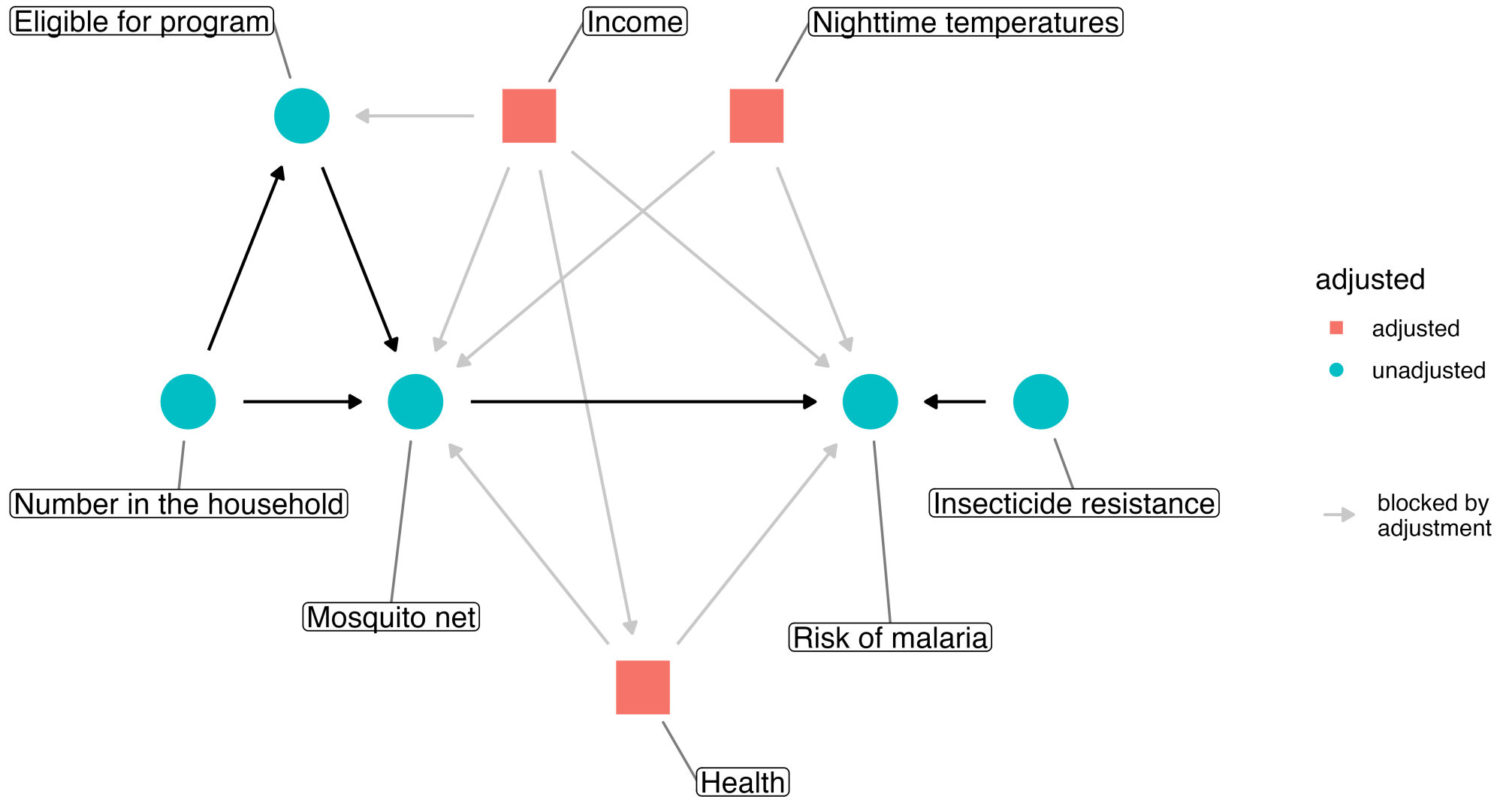
What do I need to control for?

One causal path, seven confounding

■ true effect ■ confounding effect



{health, income, temperature}



Multivariable regression: what's the association?

```
1 lm(  
2   malaria_risk ~ net + income + health + temperature,  
3   data = net_data  
4 ) |>  
5 tidy(conf.int = TRUE) |>  
6 filter(term == "netTRUE")
```

```
# A tibble: 1 × 7  
  term      estimate std.error statistic    p.value conf.low  
  <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>  
1 netTRUE    -12.0      0.314    -38.2 2.69e-232    -12.6  
# i 1 more variable: conf.high <dbl>
```

model your assumptions

**counterfactual: what if everyone used a net
vs. what if no one used a net**

Fit propensity score model

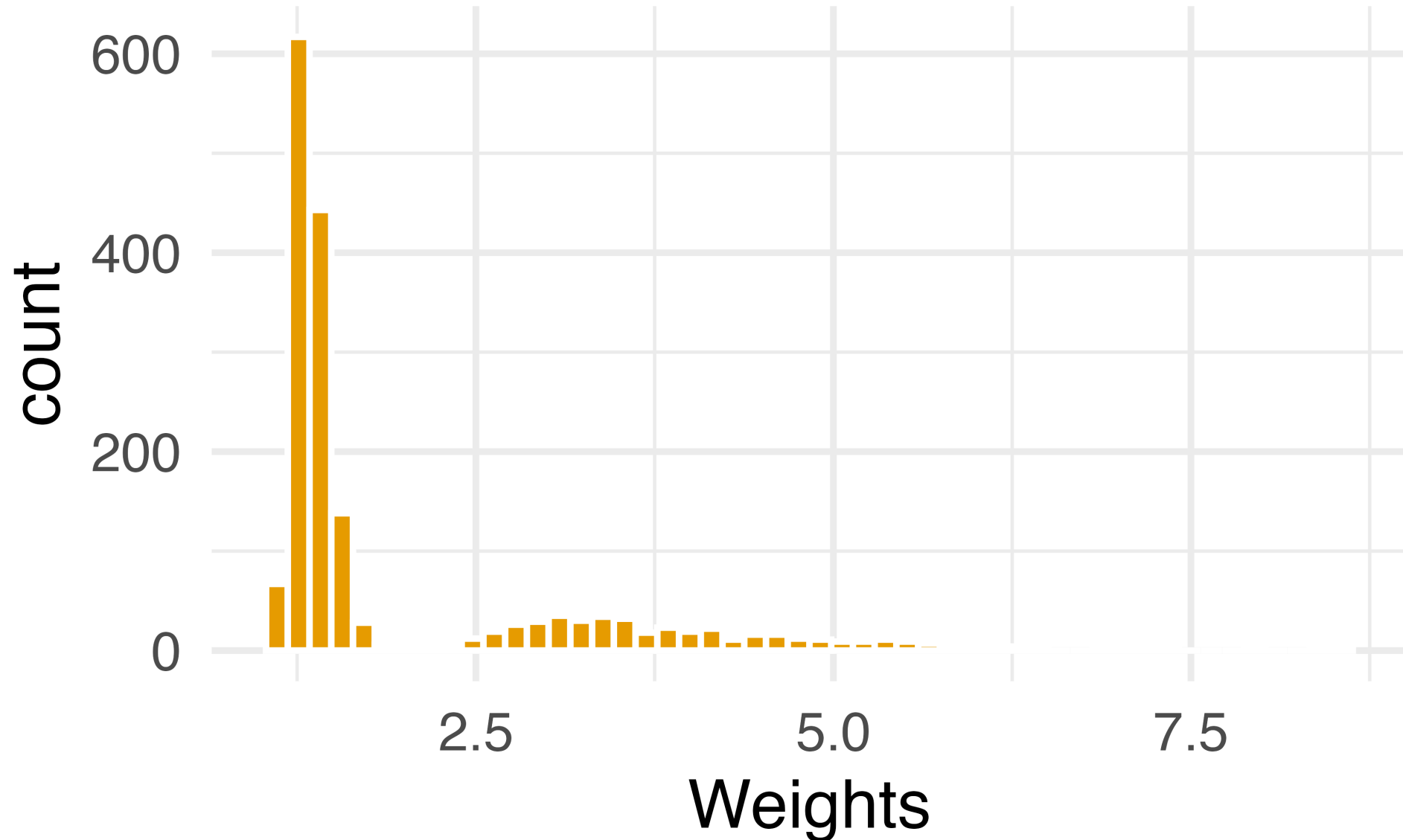
```
1 propensity_model <- glm(  
2   net ~ income + health + temperature,  
3   family = binomial(),  
4   data = net_data  
5 )
```

Calculate inverse probability weights

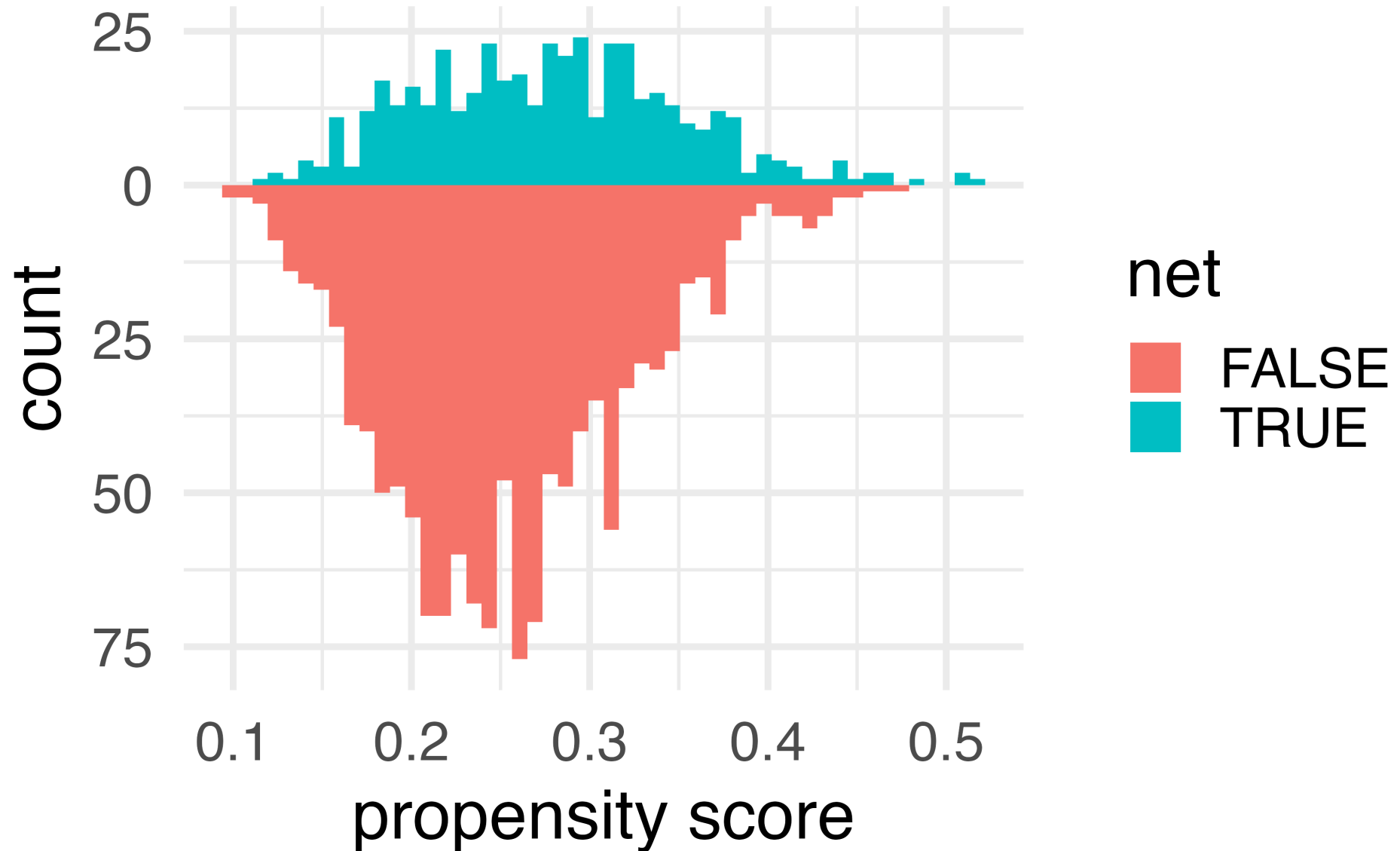
```
1 library(propensity)
2 net_data_wts <- propensity_model |>
3   # predict whether a household used a net
4   augment(data = net_data, type.predict = "response") |>
5   # calculate inverse probability
6   mutate(wts = wt_ate(.fitted, net))
```

diagnose your model assumptions

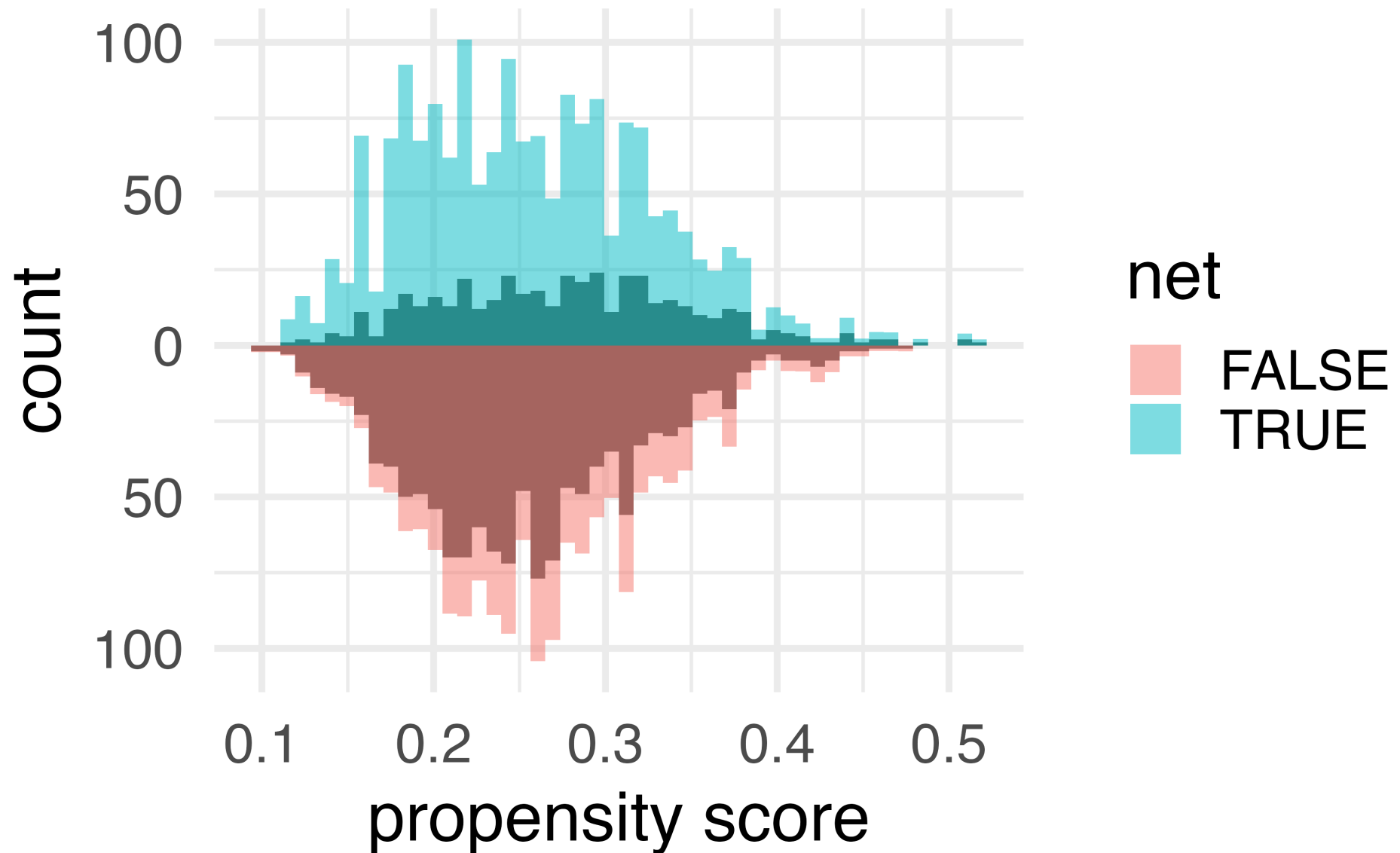
What's the distribution of weights?

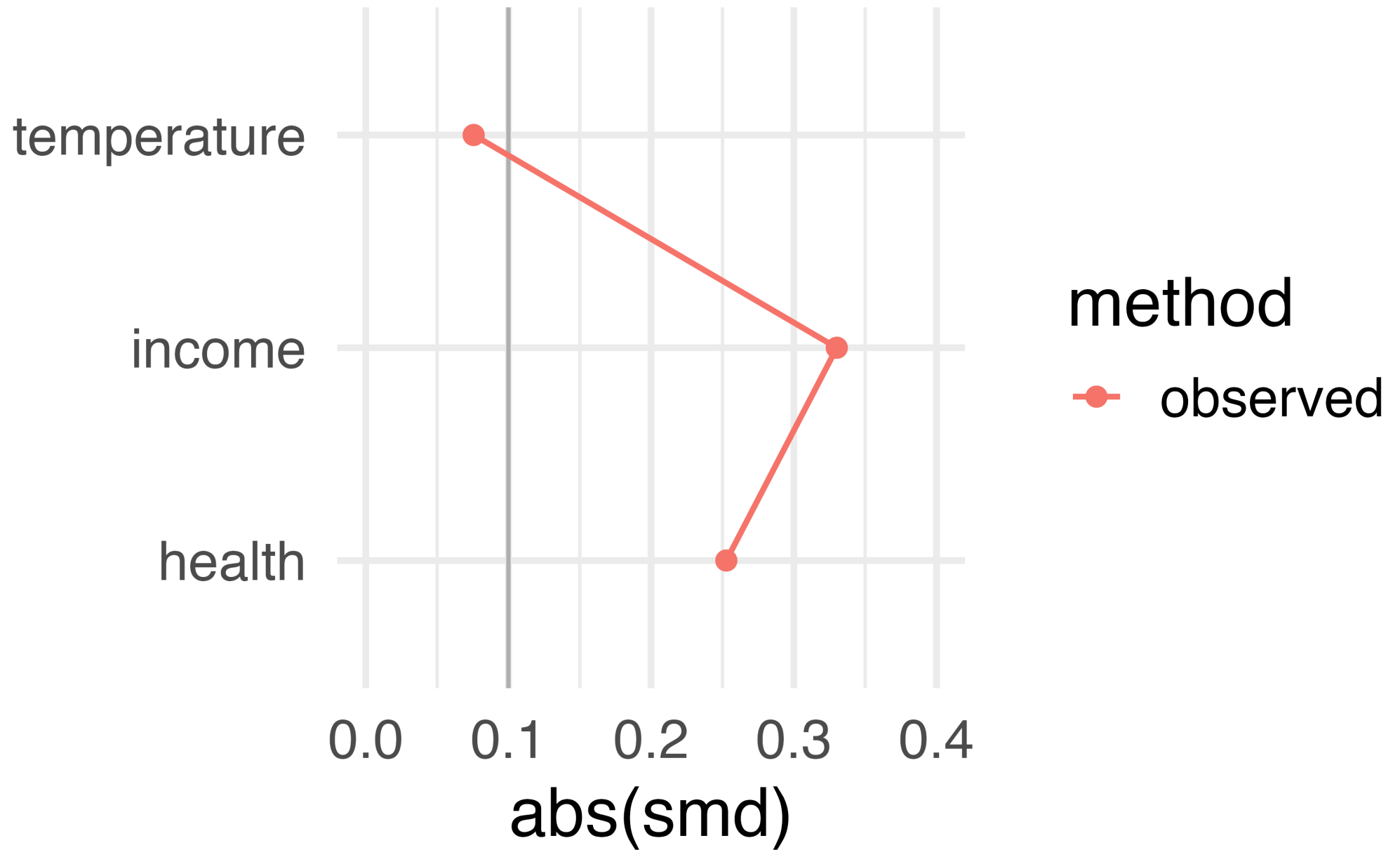


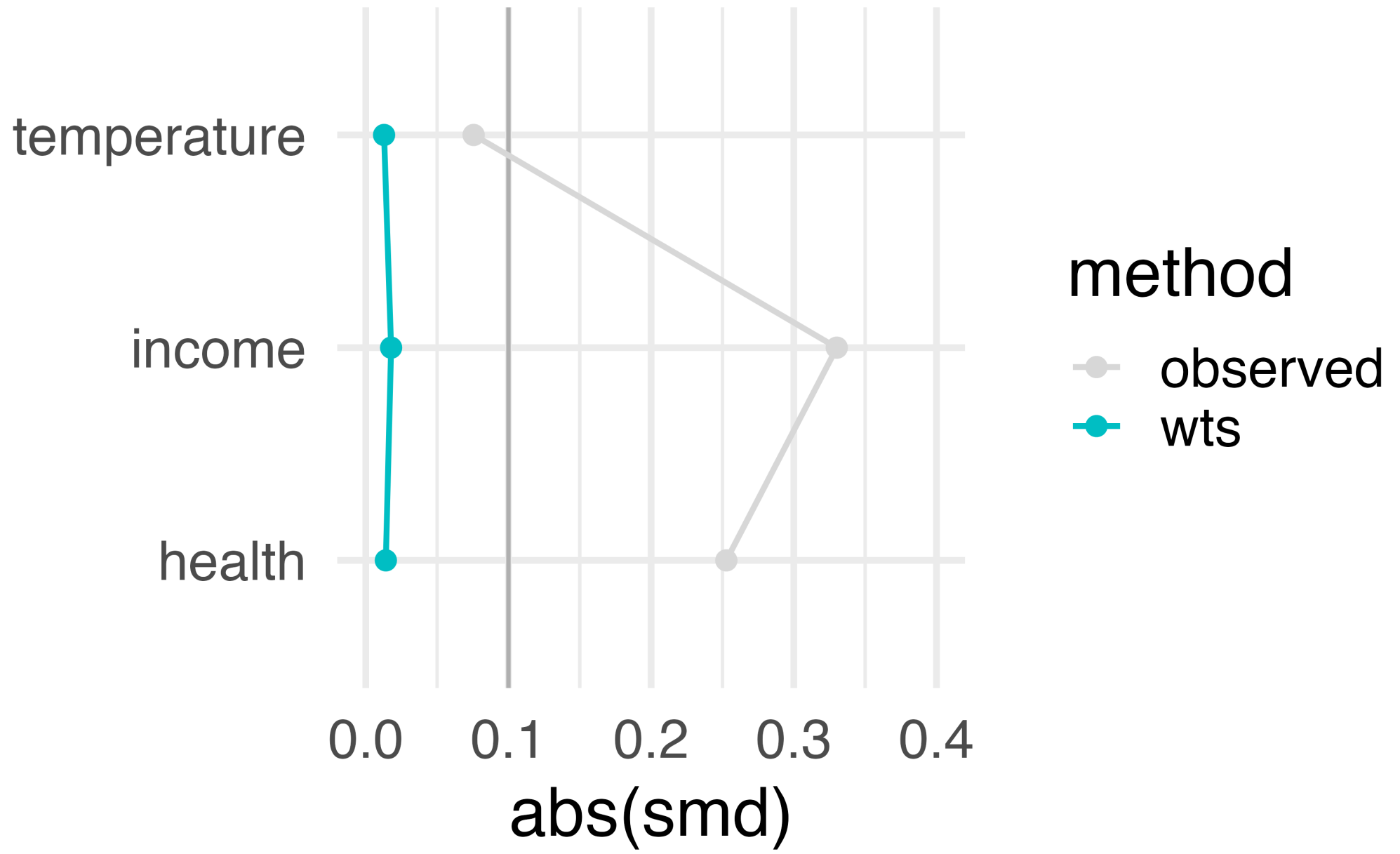
What are the weights doing?



What are the weights doing?







estimate the causal effects

Estimate causal effect with IPW

```
1 ipw_model <- lm(  
2   malaria_risk ~ net,  
3   data = net_data_wts,  
4   weights = wts  
5 )  
6  
7 ipw_estimate <- ipw_model |>  
8   tidy(conf.int = TRUE) |>  
9   filter(term == "netTRUE")
```

Estimate causal effect with IPW

```
1 ipw_estimate
```

```
# A tibble: 1 × 7
  term      estimate std.error statistic  p.value conf.low
<chr>      <dbl>      <dbl>      <dbl>    <dbl>    <dbl>
1 netTRUE    -12.5        0.624    -20.1 5.50e-81    -13.8
# i 1 more variable: conf.high <dbl>
```

Let's fix our confidence intervals with robust SEs!

```
1 # also see robustbase, survey, gee, and others
2 library(estimatr)
3 ipw_model_robust <- lm_robust(
4   malaria_risk ~ net,
5   data = net_data_wts,
6   weights = wts
7 )
8
9 ipw_estimate_robust <- ipw_model_robust |>
10   tidy(conf.int = TRUE) |>
11   filter(term == "netTRUE")
```

Let's fix our confidence intervals with robust SEs!

```
1 as_tibble(ipw_estimate_robust)
```

```
# A tibble: 1 × 9
  term      estimate std.error statistic  p.value conf.low
<chr>      <dbl>      <dbl>      <dbl>    <dbl>    <dbl>
1 netTRUE    -12.5        0.757     -16.6 2.36e-57    -14.0
# i 3 more variables: conf.high <dbl>, df <dbl>,
#   outcome <chr>
```

Let's fix our confidence intervals with the bootstrap!

```
1 # fit ipw model for a single bootstrap sample
2 fit_ipw_not_quite_rightly <- function(.split, ...) {
3   # get bootstrapped data frame
4   .df <- as.data.frame(.split)
5
6   # fit ipw model
7   lm(malaria_risk ~ net, data = .df, weights = wts) |>
8     tidy()
9 }
```

```

1 fit_ipw <- function(.split, ...) {
2   # get bootstrapped data frame
3   .df <- as.data.frame(.split)
4
5   # fit propensity score model
6   propensity_model <- glm(
7     net ~ income + health + temperature,
8     family = binomial(),
9     data = .df
10  )
11
12  # calculate inverse probability weights
13  .df <- propensity_model |>
14    augment(type.predict = "response", data = .df) |>
15    mutate(wts = wt_ate(.fitted, net))
16
17  # fit correctly bootstrapped ipw model
18  lm(malaria_risk ~ net, data = .df, weights = wts) |>
19    tidy()
20 }

```

Using {rsample} to bootstrap our causal effect

```
1 # fit ipw model to bootstrapped samples
2 ipw_results <- bootstraps(net_data, times = 1000, apparent = TRUE) |>
3   mutate(results = map(splits, fit_ipw))
```


Using {rsample} to bootstrap our causal effect

```
1 # get t-statistic-based CIs
2 boot_estimate <- int_t(ipw_results, results) |>
3   filter(term == "netTRUE")
4
5 boot_estimate
```

Using {rsample} to bootstrap our causal effect

```
# A tibble: 1 × 6
  term      .lower .estimate .upper .alpha .method
<chr>    <dbl>    <dbl>  <dbl>  <dbl> <chr>
1 netTRUE -13.4      -12.5  -11.6   0.05 student-t
```

Let's fix our confidence intervals with `ipw()`!

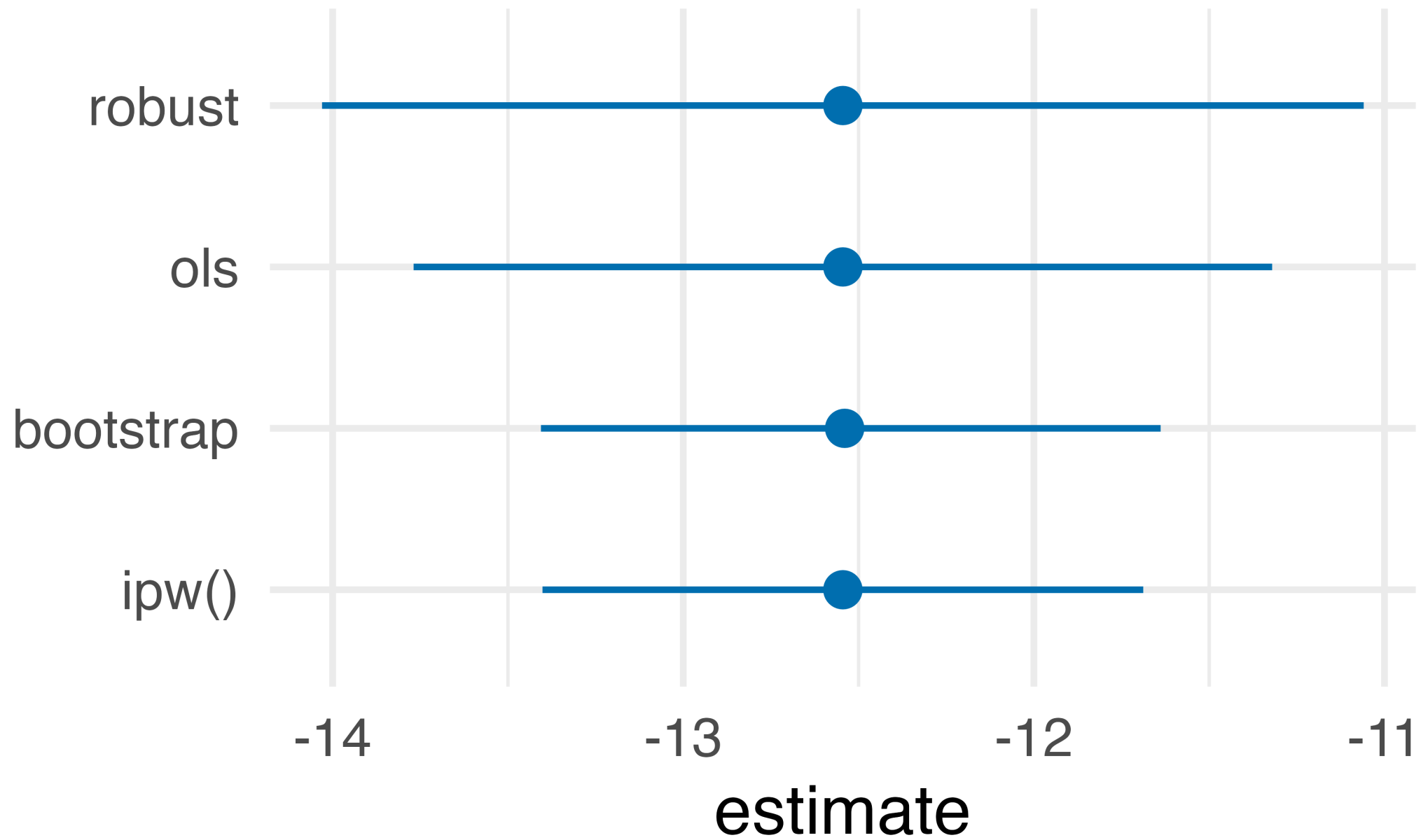
- `ipw()` uses a sandwich estimator that also accounts for the uncertainty in the propensity score model
- Bring your own models: you fit the propensity score and outcome models yourself, then supply them to `ipw()`

```
1 ipw_fit <- ipw(propensity_model, ipw_model)
```

Let's fix our confidence intervals with ipw()!

```
1 tidy(ipw_fit, conf.int = TRUE)

# A tibble: 1 × 7
  term      estimate std.error statistic p.value conf.low
<chr>    <dbl>      <dbl>      <dbl>   <dbl>   <dbl>
1 diff      -12.5      0.437      -28.7     0      -13.4
# i 1 more variable: conf.high <dbl>
```



conduct a sensitivity analysis

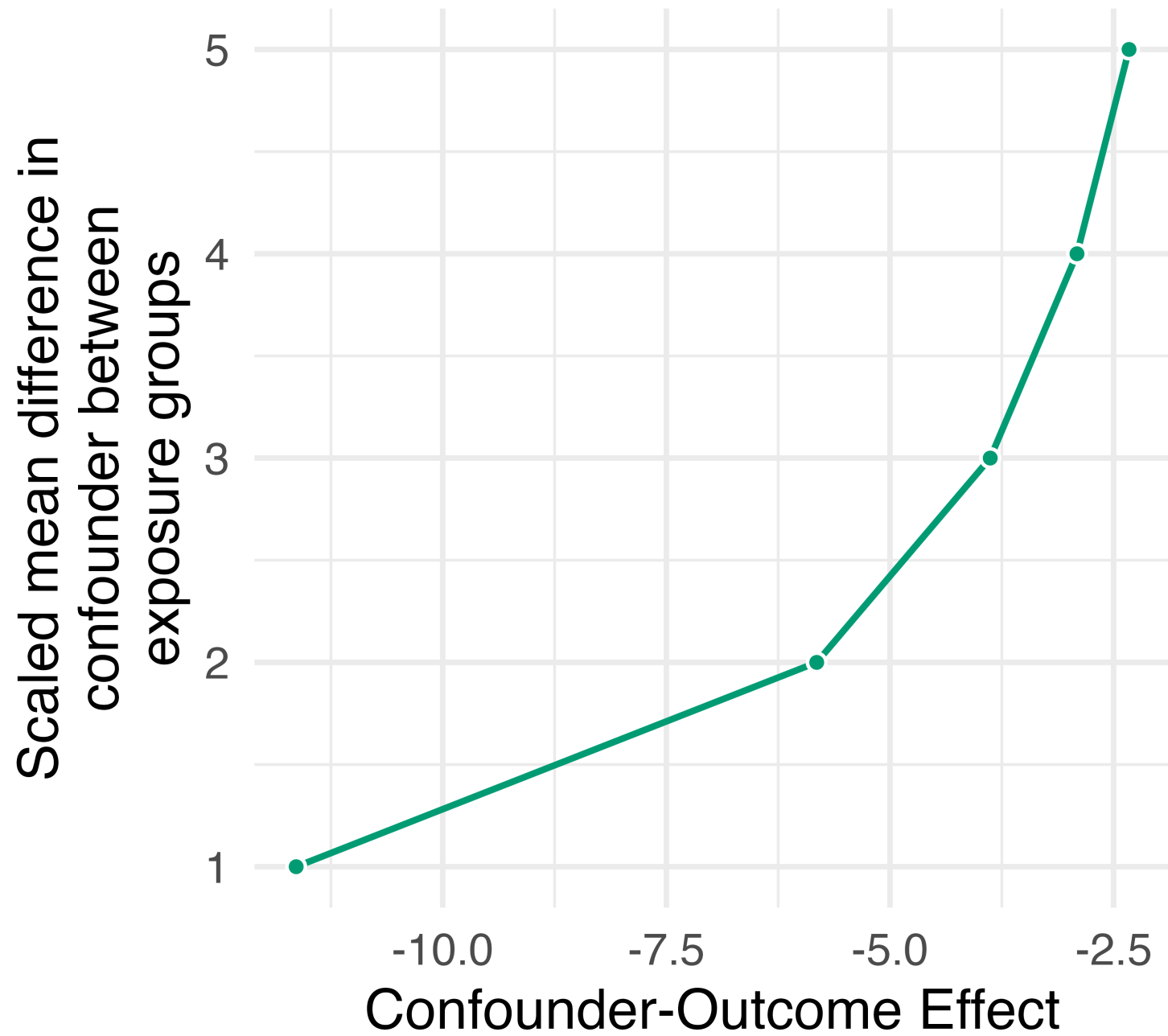
Spoiler alert: the answer we just calculated is wrong

How much confounding would tip our estimate?

```
1 library(tipr)
2 tipping_points <- tip_coef(
3   boot_estimate$.upper,
4   exposure_confounder_effect = 1:5
5 )
6
7 tipping_points |>
8   select(exposure_confounder_effect, confounder_outcome_effect)
```


How much confounding would tip our estimate?

```
# A tibble: 5 × 2
  exposure_confounder_effect confounder_outcome_effect
      <int>                <dbl>
1         1             -11.6
2         2              -5.82
3         3              -3.88
4         4              -2.91
5         5              -2.33
```



What would it take to explain away our estimate?

- A confounder with a scaled mean difference of 1 between net users and non-users would need an effect of -11.6 on malaria risk
- If the scaled mean difference were 5, an effect of -2.3 would be enough

An unmeasured confounder: genetic resistance to malaria

- 1 People with this genetic resistance have a malaria risk lower by about 10**
- 2 About 26% of people who use nets have this genetic resistance**
- 3 About 5% of people who don't use nets have this genetic resistance**

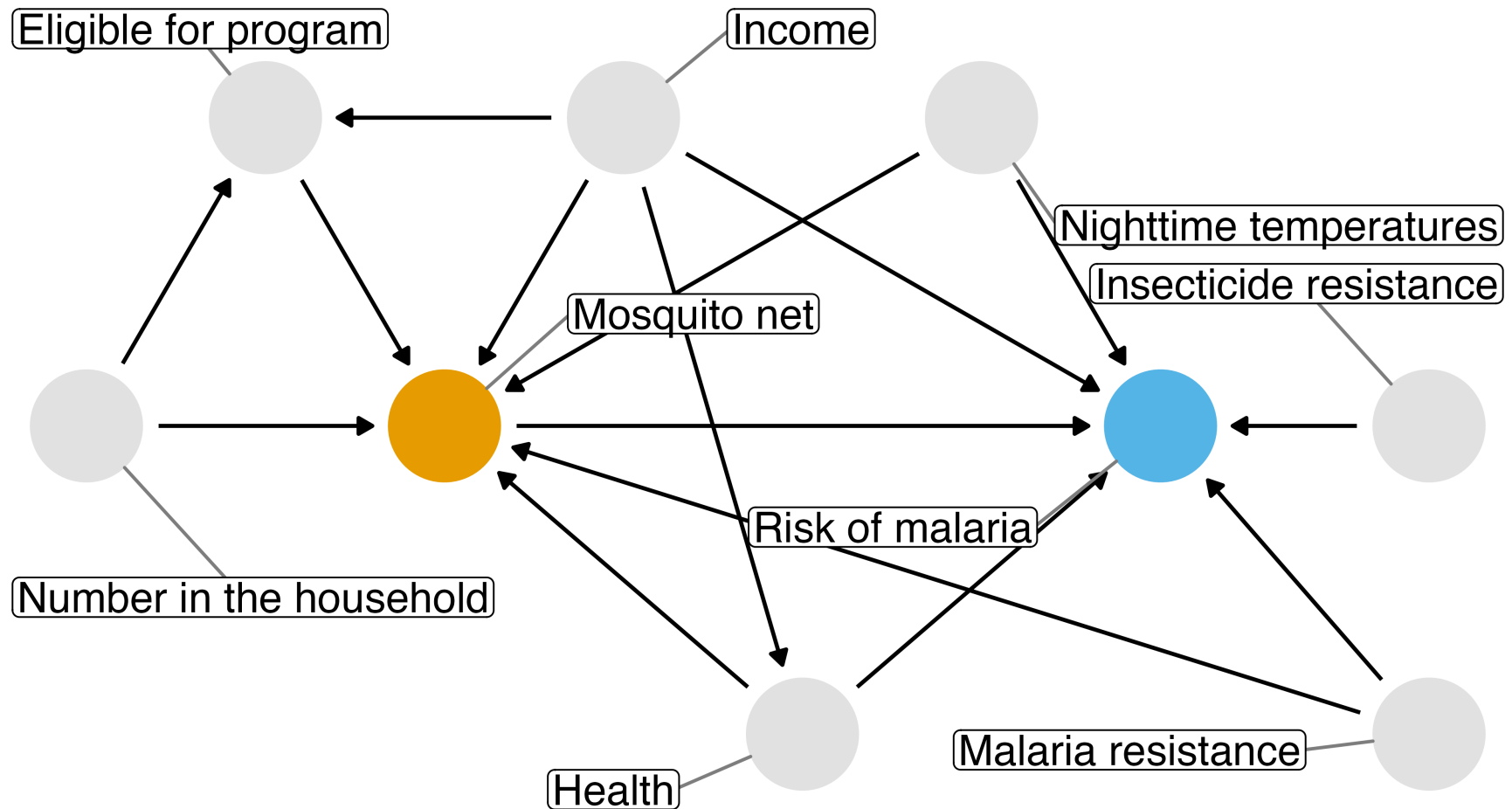
Adjust the estimate for the unmeasured confounder

```
1 adjusted_estimates <- boot_estimate |>
2   select(.estimate, .lower, .upper) |>
3   unlist() |>
4   adjust_coef_with_binary(
5     exposed_confounder_prev = 0.26,
6     unexposed_confounder_prev = 0.05,
7     confounder_outcome_effect = -10
8   )
9
10 adjusted_estimates |>
11   select(effect_observed, effect_adjusted)
```

Adjust the estimate for the unmeasured confounder

```
# A tibble: 3 × 2
  effect_observed effect_adjusted
      <dbl>         <dbl>
1      -12.5        -10.4
2      -13.4        -11.3
3      -11.6        -9.54
```

The DAG that generated the data



Re-estimate the effect with the confounder measured

```
1 # add genetic resistance to the propensity score model
2 propensity_model_full <- glm(
3   net ~ income + health + temperature + genetic_resistance,
4   family = binomial(),
5   data = net_data_full
6 )
7
8 net_data_full_wts <- propensity_model_full |>
9   augment(type.predict = "response", data = net_data_full) |>
10  mutate(wts = wt_ate(.fitted, net))
```


Re-estimate the effect with the confounder measured

```
1 ipw_model_full <- lm(  
2   malaria_risk ~ net,  
3   data = net_data_full_wts,  
4   weights = wts  
5 )  
6  
7 ipw_fit_full <- ipw(propensity_model_full, ipw_model_full)
```

Re-estimate the effect with the confounder measured

```
1 tidy(ipw_fit_full, conf.int = TRUE)
```

```
# A tibble: 1 × 7
```

	term	estimate	std.error	statistic	p.value	conf.low
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	diff	-10.2	0.469	-21.8	0	-11.2

```
# i 1 more variable: conf.high <dbl>
```

These data are simulated. The true effect of net use on malaria risk is about -10

Our causal effect estimate: -10.2 (95% CI -11.2, -9.3)

Review the Quarto file... later!

Resources

Causal Inference in R: Our book, where this example comes from. Free online.

Causal Inference: Comprehensive text on causal inference. Free online.

Bootstrap confidence intervals with {rsample}

R-causal: Our GitHub org with R packages and examples